ADVIK ESWARAN

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EDUCATION

Princeton University Princeton, NJ
Bachelor of Arts (A.B.), Geosciences *(Specialization in Geophysics)* Expected June 2027
Minors: Applied and Computational Mathematics, Computer Science

RESEARCH EXPERIENCE

University of Washington, Department of Earth & Space Sciences
Student Researcher June 2022 – Present
Advisors: Dr. T.J. Fudge, Dr. Olivia Truax

- Developed a novel Ensemble Kalman Filter methodology for Antarctic climate reconstruction, incorporating ice cores and climate models
- Reconstructed 19th-20th century Antarctic snow accumulation and sea level contribution
- Performed the first systematic, quantitative analysis of optimal Antarctic shallow ice core drilling locations
- Presented case studies for drilling new ice cores at Crary Ice Rise and Lyddan Ice Rise, and redrilling the ITASE cores in West Antarctica
- Analyzed recent changes in Antarctic surface mass balance in the context of natural variability, and compared to ice sheet model projections

Ice Core Extension to Today's Observational Period (ICE-TOP)
Student Researcher June 2026 – Present
Advisors: Dr. Holly Winton, Dr. Olivia Truax

- I was invited to join ICE-TOP, an international research collaboration between researchers in New Zealand, France, and the USA. The goal of ICE-TOP is to identify the best ice core sites to redrill for the purpose of quantifying polar amplification and evaluating ice sheet model projections.
- I developed an optimal sensor-based technique, extending on previous work, to optimize the selection of ice core sites for this goal, finding that the most optimal sites were located in coastal West Antarctica

Princeton University / Center for Oldest Ice Exploration
Student Researcher January 2026 – Present
Advisor: Dr. John Higgins

- Ran Community Ice Sheet Model simulations on multi-millennial year timescales, in order to identify optimal ice core drilling locations for obtaining the oldest ice.
- Developing a method to empirically model folding, thinning, and diffusion in blue ice cores from the Allan Hills, Antarctica.

National Centre for Atmospheric Science

Student Researcher June 2025 – August 2025

Advisor: Dr. Ryan Neely III

- Produced 2D spatial reconstructions of Antarctic snow accumulation using atmospheric radar profiles
- Developed a process-based snowfall retrieval scheme from a scanning Doppler radar at Troll Station, Antarctica

Juneau Icefield Research Program

Student Researcher June 2024 – August 2024

Advisors: Dr. Eric Steig, Dr. Bradley Markle

- Contributed to numerous field glaciology research projects on the Juneau Icefield, including the drilling and processing of three ~15 m ice cores for water isotope sampling and installation of a TRI radar station to study supraglacial hydrology

PUBLICATIONS

Eswaran, A., Truax, O. J., MacDonell, S., Matsuoka, K., Neff, P. D., Steig, E.J., Winton, V.H.L., & Fudge, T. J. (2026). Data-Driven Optimization of Antarctic Shallow Ice Core Site Selection. *Journal of Glaciology*, 72, e102. doi:10.1017/jog.2026.10218

Eswaran, A., Truax, O.J., & Fudge, T.J. (2024). 20th-Century Antarctic Sea Level Mitigation Driven by Uncertain East Antarctic Accumulation History. *Geophysical Research Letters,* 51, e2023GL106991. https://doi.org/10.1029/2023GL106991

In Preparation

Eswaran, A., Truax, O.J., & Fudge, T.J. (2026). 21st-Century Increases in Antarctic Surface Mass Balance Exceed Natural Variability. (Manuscript in Preparation).

CONFERENCE PRESENTATIONS

Eswaran, A., Truax, O.J., MacDonell, S., Matsuoka, K., Neff, P.D., Steig, E.J., Winton, V.H.L., & Fudge, T.J. (2026). Data-Driven Optimization of Antarctic Shallow Ice Core Site Selection. International Partnerships in Ice Core Sciences, Banff, AB, Canada, 11-16 Oct. (Poster)

Winton, V.H.L., Casado, M., Eswaran, A., Harris, M., Agosta, C., Amory, C., Clem, K., Katurji, M., Truax, O.J. (2026). ICE-TOP: Ice Core Extension to Today's Observational Period, International Partnerships in Ice Core Sciences, Banff, AB, Canada, 11-16 Oct. (Poster, Presenting Author)

Eswaran, A., Truax, O.J., MacDonell, S., Matsuoka, K., Neff, P.D., Steig, E.J., Winton, V.H.L., & Fudge, T.J. (2026). ITASE Redrilling: Bridging the Gap Between Paleoclimatic and Modern Accumulation Variability in Antarctica. 33rd WAIS Workshop, Front Royal, VA, 28-30 Sep. (Oral)

Eswaran, A., Truax, O., & Fudge, T.J. (2025), Optimal Sensor Analysis for Antarctic Ice Core Site Identification. 4th US Ice Core Community Meeting, Minneapolis, MN, 12-14 May. (Oral)

Fudge, T.J., Eswaran, A., & Truax, O. (2024), Optimal Sensor Analysis for Identification of Antarctic Ice Core Sites, C11C-0492 presented at 2024 Fall Meeting, AGU, Washington, DC, 9-13 Dec. (Poster, Presenting Author)

Eswaran, A., Truax, O., & Fudge, T.J. (2024), Antarctic 20th-Century Sea Level Mitigation Dominated By Uncertainty in 19th-Century East Antarctic Snow Accumulation, EGU24-14163 presented at EGU General Assembly 2024, Vienna, Austria, 14-19 Apr. (Oral)

Eswaran, A., Truax, O., & Fudge, T.J. (2023), Late 20th-century Increase in Antarctic Snow Accumulation Drives Modest Mitigation of Sea Level Rise, C33E-1455 presented at 2023 Fall Meeting, AGU, San Francisco, CA, 11-15 Dec. (Poster)

Eswaran, A., Truax, O., & Fudge, T.J. (2023). Late 20th-century Increase in Antarctic Snow Accumulation Drives Modest Mitigation of Sea Level Rise. Northwest Glaciologists Meeting, Seattle, WA, 20-21 October. (Oral)

Eswaran, A. and Fudge, T.J. (2023), Reconstructing Antarctic Accumulation With Data Assimilation. 2nd US Ice Core Community Meeting, Seattle, WA, 8-10 May. (Poster)

Eswaran, A. and Fudge, T.J. (2022), Antarctic Ice Core Site Identification Using Data Assimilation of Synthetic Proxy Records for Twentieth Century Climate Reconstruction, C32D-0873 presented at 2022 Fall Meeting, AGU, Chicago, IL, 12-16 Dec. (Poster)

INVITED TALKS

"Optimizing Antarctic Ice Core Site Selection to Constrain Modern Climate Variability", ICE-TOP Workshop, International Partnerships in Ice Core Sciences, Banff, AB, Canada, 11 October 2026.

"Optimal Sensor Analysis for Antarctic Shallow Ice Core Site Identification", British Antarctic Survey, Cambridge, United Kingdom, 3 July 2025.

"Antarctic 20th-Century Sea Level Mitigation Dominated By Uncertainty in 19th-Century East Antarctic Snow Accumulation", International Glaciological Society Global Seminar, 9 October 2024.

GRANTS/AWARDS

Research Accelerator Grant, University of Washington Program on Climate Change, 2026

- Project funding for "Contextualizing Recent Antarctic Surface Mass Balance Change Within Paleoclimatic Natural Variability"

Student-Initiated Internship Award, High Meadows Environmental Institute, Princeton University, 2026

- Summer research visit funding to the University of Canterbury, Christchurch, New Zealand

Research Accelerator Grant, University of Washington Program on Climate Change, 2025

- Project funding for "Shallow Ice Core Drilling at Crary Ice Rise: A Pilot Project Proposal"

Student-Initiated Internship Award, High Meadows Environmental Institute, Princeton University, 2025

- Summer research visit funding to the National Centre for Atmospheric Science, Leeds, United Kingdom

Research Accelerator Grant, University of Washington Program on Climate Change, 2024

- Project funding for "Optimal Sensor Analysis for Identifying Shallow Antarctic Ice Core Drilling Locations"

Summer Undergraduate Research Award, NASA Space Grant, 2023

- Summer research award from the Washington NASA Space Grant Consortium

Outstanding Student Presentation Award, AGU Fall Meeting, 2022

- Chosen as one of the top six student presenters in the Cryosphere section

Summer Undergraduate Research Award, NASA Space Grant, 2022

- Summer research award from the Washington NASA Space Grant Consortium

Full Membership, Sigma Xi

SERVICE

Reviewer for *Geophysical Research Letters*

Reviewer for *The Cryosphere*

Co-Reviewer for *Nature Communications*